

Several Complex Variables

Labix

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1.1

Definition 1.1.1 (Holomorphic Functions of Several Variables)

Let $U \subseteq \mathbb{C}^n$ be open. Let $f = (f_1, \dots, f_n) : U \rightarrow \mathbb{C}$ be a function. Let $p \in U$. We say that f is holomorphic at p if the Cauchy-Riemann condition is satisfied:

$$\frac{\partial f}{\partial \bar{z}_k} = \frac{1}{2} \left(\frac{\partial f}{\partial x_k} - \frac{1}{i} \frac{\partial f}{\partial y_k} \right) = 0$$

for $1 \leq k \leq n$.

Definition 1.1.2 (Polydiscs)

Let $(a_1, \dots, a_n) \in \mathbb{C}^n$. Let $r_i > 0$ for $1 \leq i \leq n$. Define the polydisc at $(a_1, \dots, a_n)$ of polyradii $(r_1, \dots, r_n)$ to be

$$\Delta_{(a_1, \dots, a_n)}(r_1, \dots, r_n) = \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid |z_i - a_i| < r_i \text{ for } 1 \leq i \leq n\}$$

Definition 1.1.3 (The Boundary Torus)

Let $(a_1, \dots, a_n) \in \mathbb{C}^n$. Let $r_i > 0$ for $1 \leq i \leq n$. Define the boundary torus of the polydisc $\Delta_{(a_1, \dots, a_n)}(r_1, \dots, r_n)$ to be

$$T_{(a_1, \dots, a_n)}(r_1, \dots, r_n) = \{(z_1, \dots, z_n) \in \mathbb{C}^n \mid |z_i - a_i| = r_i \text{ for } 1 \leq i \leq n\}$$

The boundary torus is not quite the boundary of the polydisc: it is strictly smaller.

Theorem 1.1.4 (Cauchy's Integral Formula)

Let $U \subseteq \mathbb{C}^n$ be open. Let $f : U \rightarrow \mathbb{C}$ be a holomorphic function. Let $\Delta_{(a_1, \dots, a_n)}(r_1, \dots, r_n) \subseteq U$. Then we have

$$f(z_1, \dots, z_n) = \frac{1}{(2\pi i)^n} \int_{T_{(a_1, \dots, a_n)}(r_1, \dots, r_n)} \frac{f(w_1, \dots, w_n)}{(w_1 - z_1) \cdots (w_n - z_n)} dw_1 \cdots dw_n$$

for all $(z_1, \dots, z_n) \in \Delta_{(a_1, \dots, a_n)}(r_1, \dots, r_n)$.

1.2 Power Series

1.3 Analytic Functions